

GABRIEL WOMELSDORF

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SUMMARY

Computer Engineering student (B.S. 2027) with hands-on experience in PCB design, embedded firmware, and power electronics, alongside 3+ years of part-time industry experience at Core4ce. Interested in embedded systems, power delivery engineering, and applied hardware research.

EDUCATION

Christopher Newport University

Newport News, VA

B.S., Computer Engineering, *expected 2027*

Minor, Mathematics

SKILLS

Languages: Python, C, C++, Java, C#, Z80 Assembly

Embedded Platforms: ESP32/ESP-IDF, Nordic nRF52/nRF Connect, ESPHome

Hardware/Systems: PCB Layout, SystemVerilog, PMBus

Frameworks/Tools: Docker, Apache NiFi, Django, Git, AWS

PERSONAL RESEARCH & PROJECTS

- **RPi CM5 ATX Carrier Board (In Progress):** Designing an 8-layer Raspberry Pi Compute Module 5 carrier board targeting standard nano-ATX form factor compatibility. Features per-rail power telemetry at 10A/rail using TPS544B27W buck converters over PMBus, with power planes and controlled impedance stackup for high-current distribution and EMI management. Developing supervisory firmware on an ESP32 for rail sequencing, fault detection, and telemetry reporting.
- **Air Sentry:** Designed and built several hardware revisions of an indoor air quality monitoring system with custom schematic capture, PCB layout, and board fabrication. Fifth revision migrated to an nRF52810 for ultra-low-power operation, achieving approximately one week of battery life. Developed custom drivers for the STCC4 CO₂ sensor and integrated the system into a networked sensor platform. Currently being used for research in the additive manufacturing space.
- **Z80 Microcomputer:** Designed an 8-bit Z80-based microcomputer and programmed a minimalist operating system in Z80 assembly supporting basic I/O and memory routines.

ENGINEERING EXPERIENCE

Core4ce

Reston, VA

Software R&D Intern → Software Engineer → Lead Software Engineer (part-time)

Jun 2022 – Present

- Designed and implemented a production data pipeline from scratch in Apache NiFi on AWS, developing custom Java processors including IAM-compliant AWS integrations and fully reconfigurable routing logic, enabling dynamic data flow configuration without code changes.
- Led a two-month effort to overhaul testing for a previously untested subsystem of Cyberscape, significantly increasing test coverage and bringing the component to a production-ready state.
- Led end-to-end development of the Cyberscape project, owning architecture decisions, sprint planning, and client-facing deployments while mentoring junior engineers.
- Expanded data forensics capabilities in Cyberscape by integrating Django services containerized with Docker and deploying a honeypot on AWS for threat research.

Christopher Newport University

Newport News, VA

Student Researcher (part-time)

Nov 2023 – May 2025

- Conducted exploratory research using CNC routing machines to fabricate table-sized PCBs for the Super BigByte project.
- Assembled, installed, and documented key power delivery systems for the Photo-Multipliers on the coordinate detector for the Super BigByte project.

LEADERSHIP & INVOLVEMENT

The Captains Workshop, CNU

President

Feb 2023 – Present

- Founded and lead a club focused on 3D printing; organized hands-on workshops for members.

Dean's Advisory Board, CNU School of Engineering

Member

Oct 2023 – Jan 2025

- Serve as liaison between students and the Dean, advocating for student needs and proposing solutions to key issues within the school.